

Code:

```
#include <stdio.h>

#define MAXN 100005

int pre[MAXN];
int in_pos[MAXN];

void printPostOrder(int pre_start, int in_start, int in_end) {
    if (in_start > in_end) return;

    int root_val = pre[pre_start];
    int root_idx = in_pos[root_val];
    int left_size = root_idx - in_start;

    // Recurse left subtree
    printPostOrder(pre_start + 1, in_start, root_idx - 1);
    // Recurse right subtree
    printPostOrder(pre_start + 1 + left_size, root_idx + 1, in_end);

    printf("%d ", root_val);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int i;
    for(i=1; i<=n; i++) {
        scanf("%d", &pre[i]);
    }
    for(i=1; i<=n; i++) {
        int val;
        scanf("%d", &val);
        in_pos[val] = i;
    }

    printPostOrder(1, 1, n);
    printf("\n");
    return 0;
}
```

Input:

```
5
5 3 2 1 4
3 5 1 2 4
```

Output:

```
3 1 4 2 5
```